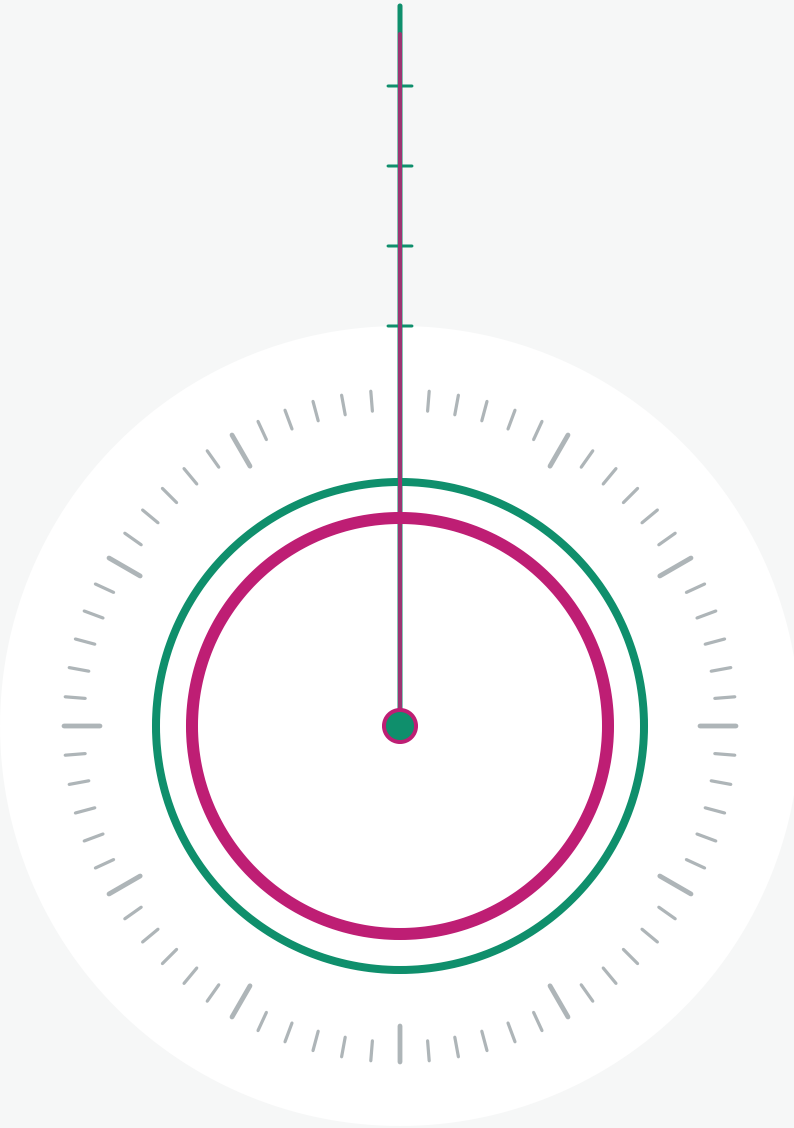
DATUM LINE

Every lab’s cells. One “normal”.

A shared map of every lab’s control cells.



Kat Titterton
Cofounder, CSO, San Francisco



macrofluidic.me/resume

8
peer-reviewed papers
two in Nature Communications ↗

2
patent applications
filed at Axiom Bio

100,000+
compounds imaged
two Cell Painting datasets ↗

Kat Titterton

WORK

2023–2026 **Axiom**
cell imaging, AI

2022–2023 **insitro**
automated screening

2021–2022 **MIT**
expansion microscopy

2018–2021 **abbvie**
neuron assays

2017–2018 **MGH**
electrophysiology

EDUCATION

2013–2017 **COLGATE** BA
neuroscience, math

MY WORK

1. Cell Painting in primary human hepatocytes
   Cell Systems 2026, Ewald, Broad Institute
2. Counting cells predicts bioactivity benchmarks
   Nat Commun 2026, Seal, Univ. of Cambridge
3. Cell Painting of hepatotoxic botanicals
   bioRxiv 2026, Ollitrault, EMBL-EBI
4. An antibody to pathological tau
   Mol Neurodegen Adv 2026, Yanamandra, AbbVie
5. Capturing synaptic proteopathic seeds
   iScience 2026, Sebastian, AbbVie
6. A neuronal tau aggregation assay
   J Biol Chem 2025, Wu, AbbVie
7. Multiplexed imaging of brain nanostructures
   Nat Commun 2024, Kang, MIT
8. Tau aggregation in human iPSC neurons
   iScience 2022, Manos, AbbVie
9. Neurosteroid photolabeling of $GABA_A$ receptors
   Eur J Med Chem 2019, Wu, U Illinois Chicago

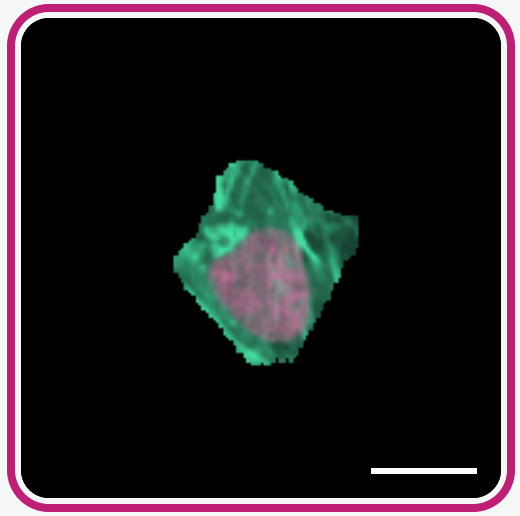
Do my cells look “normal”?

“normal” = published images + past data



asked @ five benches
... answered @ none

“Normal” compared with whom?
Each lab’s own checks can pass while labs still disagree (slides 03 and 04, archive A13).



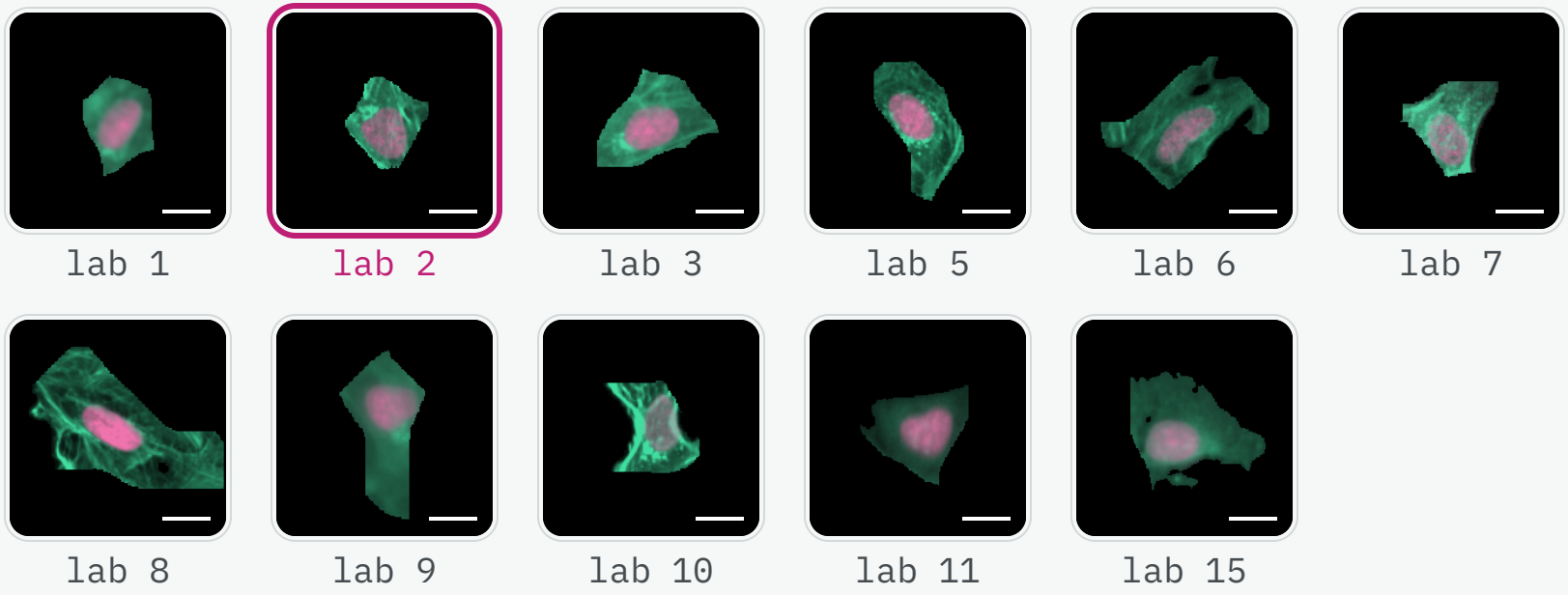
my cells, lab 2

one DMSO cell, 20 µm bar

Each lab judges “normal” alone.

$28B a year of US lab research others cannot reproduce [6]

36% ≈ $10B, cells and reagents | 64%, other causes



nuclei (DNA) cell body (actin, membrane) — 20 µm

JUDGING ALONE COSTS

$100,000s

a screen run on cells that changed

[28]: about $50k a run, 3 runs a week

months

asking around, or re-running to be sure [28]

One lab, or eleven?

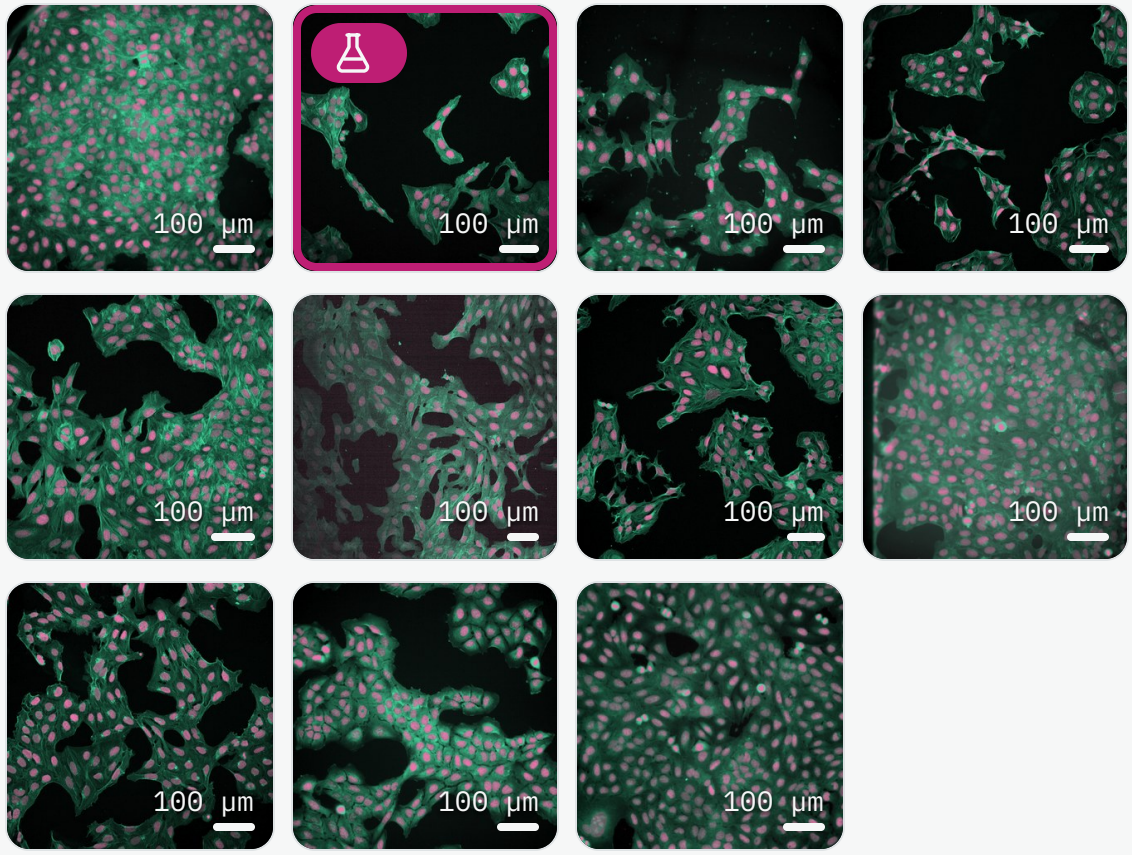


fig 1 Cell Painting, images: DNA channel, classifier: 3,773 numbers per well from all channels, 93,228 wells, 11 labs (a lab = one JUMP data source, its data-generating centre), plates held out, bars 100 µm, JUMP [7], archive A2, A1, archive A5

same cells

U2OS: human bone cancer cells

osteosarcoma (bone cancer) from a 15-year-old’s tibia, described in 1967 [26]

the one line in JUMP’s full dataset, so every lab here grew it [27]

chosen over A549 by a slim margin, for its public data [27]

named in 11 of the Cell Painting Gallery’s 28 datasets [17]

a cancer line: “normal” means its DMSO baseline, not healthy tissue

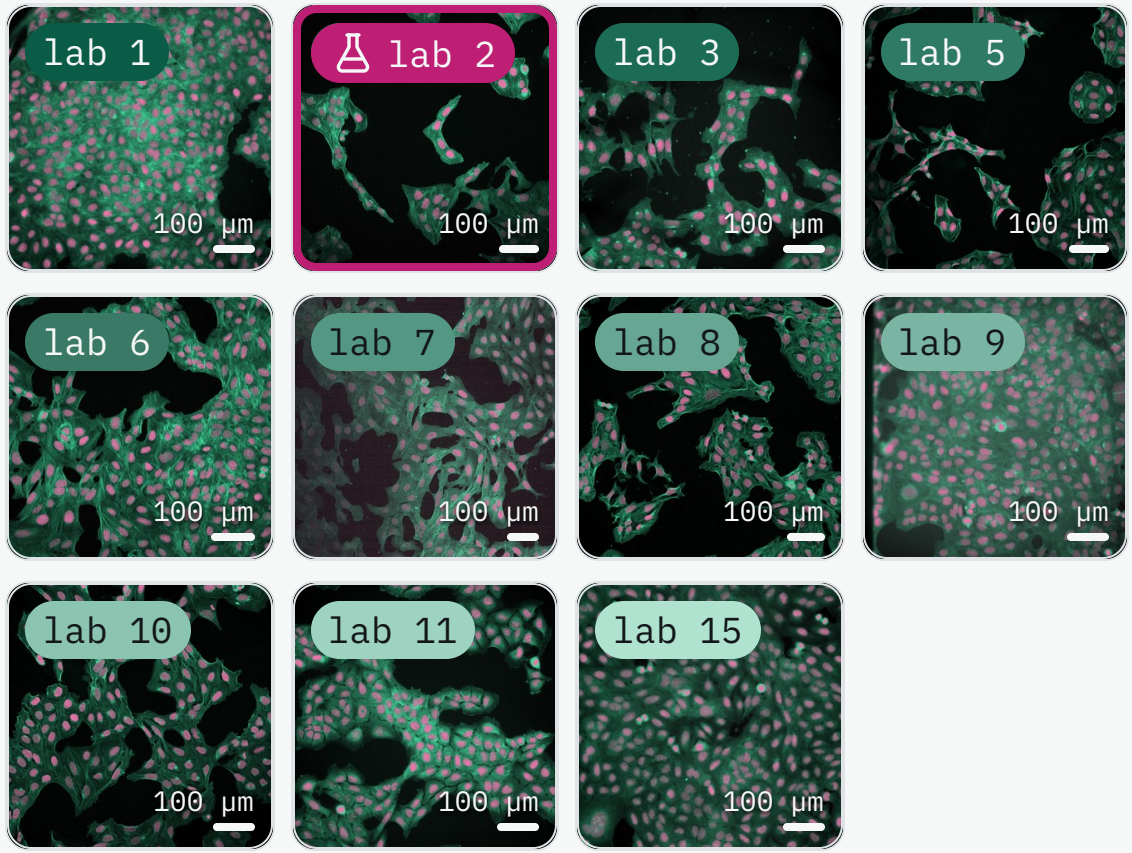
same protocol

Cell Painting, one protocol at every site [7, 27]

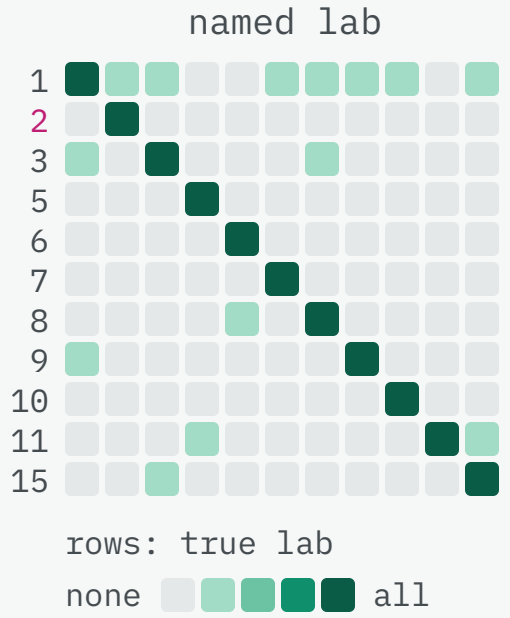
no drug

DMSO only, the drugs’ solvent [7]

Could anyone tell?
A classifier can.



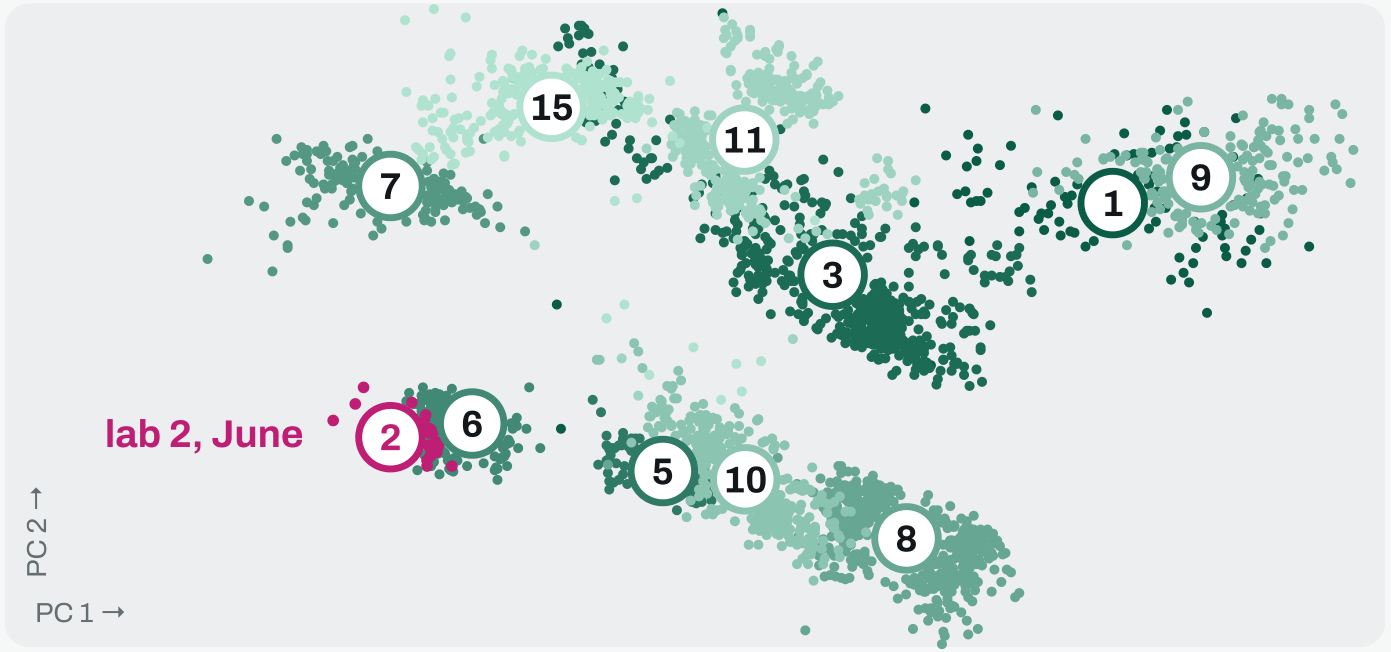
99.9% right lab, of 11
chance: 1 in 11



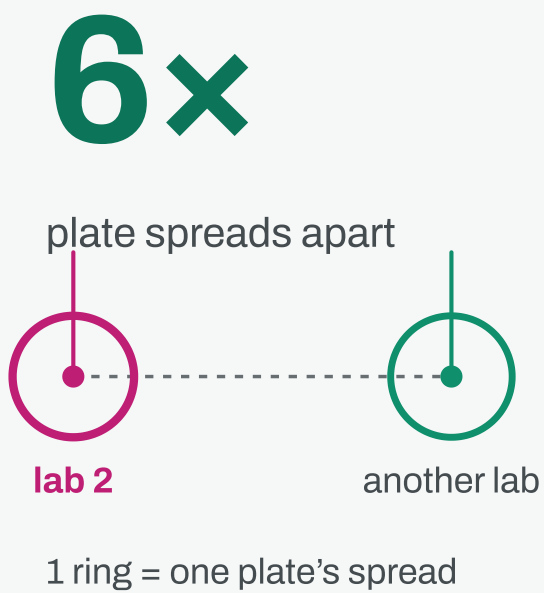
Biosecurity: labels can lie, cells cannot.
Good actors can prove what they test. A8

fig 1 Cell Painting, images: DNA channel, classifier: 3,773 numbers per well from all channels, 93,228 wells, 11 labs (a lab = one JUMP data source, its data-generating centre), plates held out, bars 100 µm, JUMP [7], archive A2, A1, archive A5

Eleven labs.
Eleven “normals”.



one well 7 a lab’s “normal” closer = more alike



dots: DMSO wells, 8 per plate, one in three drawn; PCA of 15,352 wells, for the eye (distances use all 3,773 numbers); plates: 384 wells, 1,536 in labs 1 and 9; 6×: two labs’ centres sit 6.2 plate spreads apart on average, a plate spread being a well’s mean distance from its own plate’s centre, all 3,773 numbers (results/e13); JUMP [7]; archive A2, A1, archive A13

Send the controls.
Know it’s “normal” before the screen.

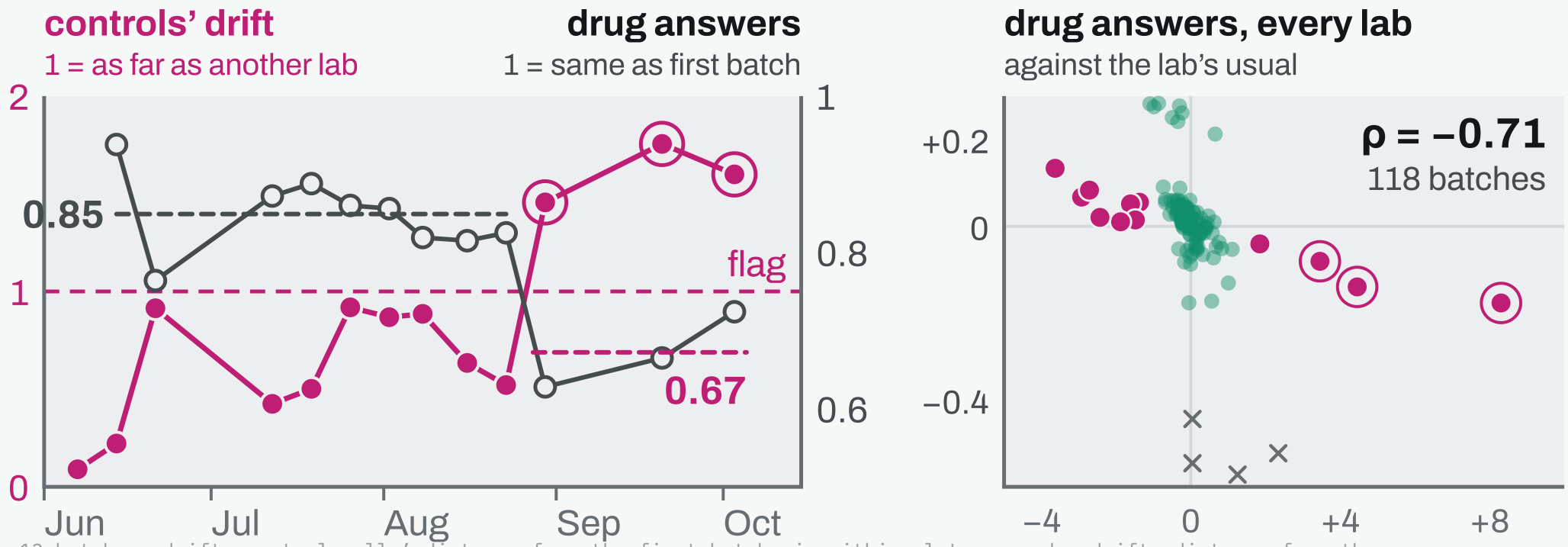


Certificate
for the new batch

- controls “normal” ✓
- drugs match ✓
- grew as usual ✓

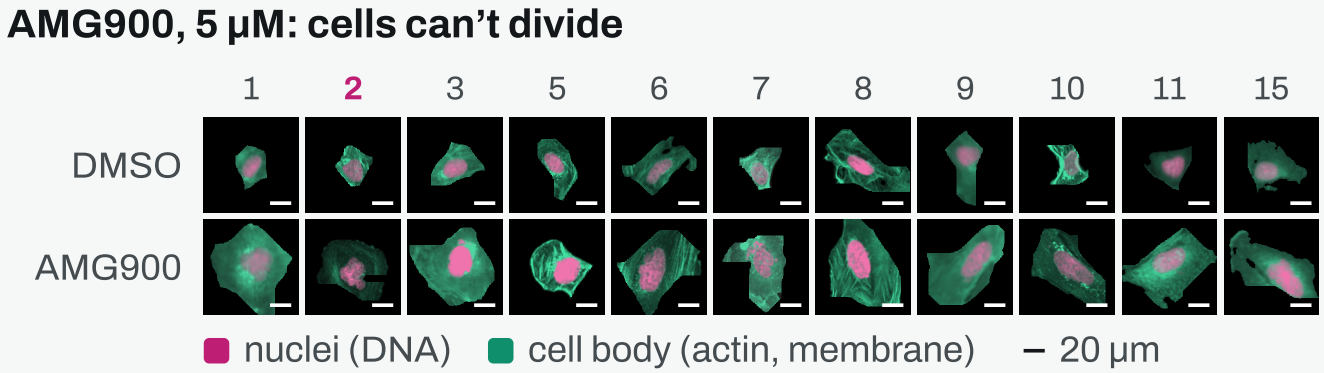
a proposed certificate, replayed on public data; closest lab in all 3,773 numbers: lab 6, 4.4 within-plate spreads away; known drugs worked; agreement with other labs 0.70 measured, 0.77 on the map; 188 cells per well, in its usual range; JUMP [7], results/e9, e11, e12; archive A11, A6

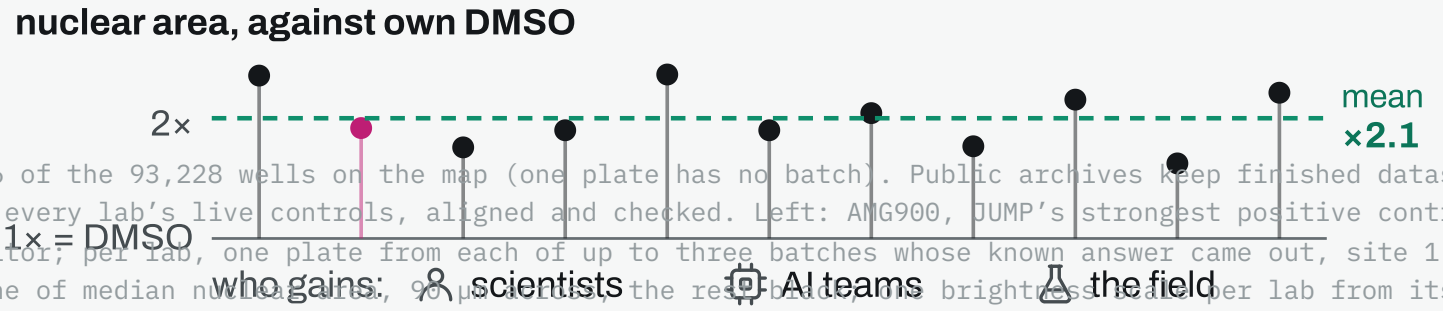
Controls catch drift before any drug is read.

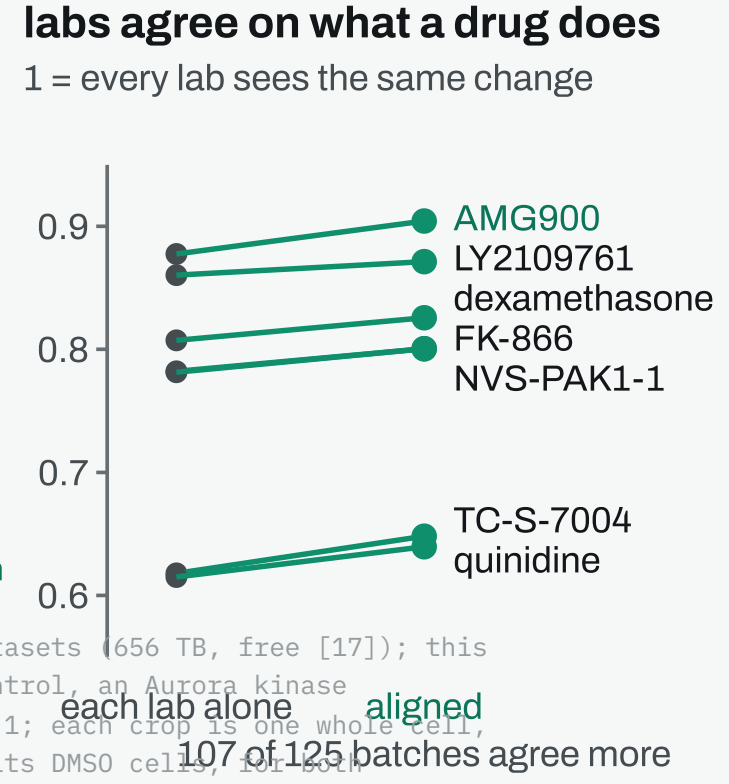


lab 2, 13 batches; drift: control wells' distance from the first batch, in within-plate spreads; drift: distance from the first batch over distance to the nearest other lab's centre, so 1 is the flag line (most plates nearer another lab than the first batch, e9); match: seven known drugs, 5 µM, against the first batch; dashed: means; right: every batch's drift and match minus its lab's mean, Spearman over all labs, permutation p = 0.0005, crosses: positive controls failed (results/e10); archive A6; JUMP [7]

The product is the data.
One drug, one shared change.

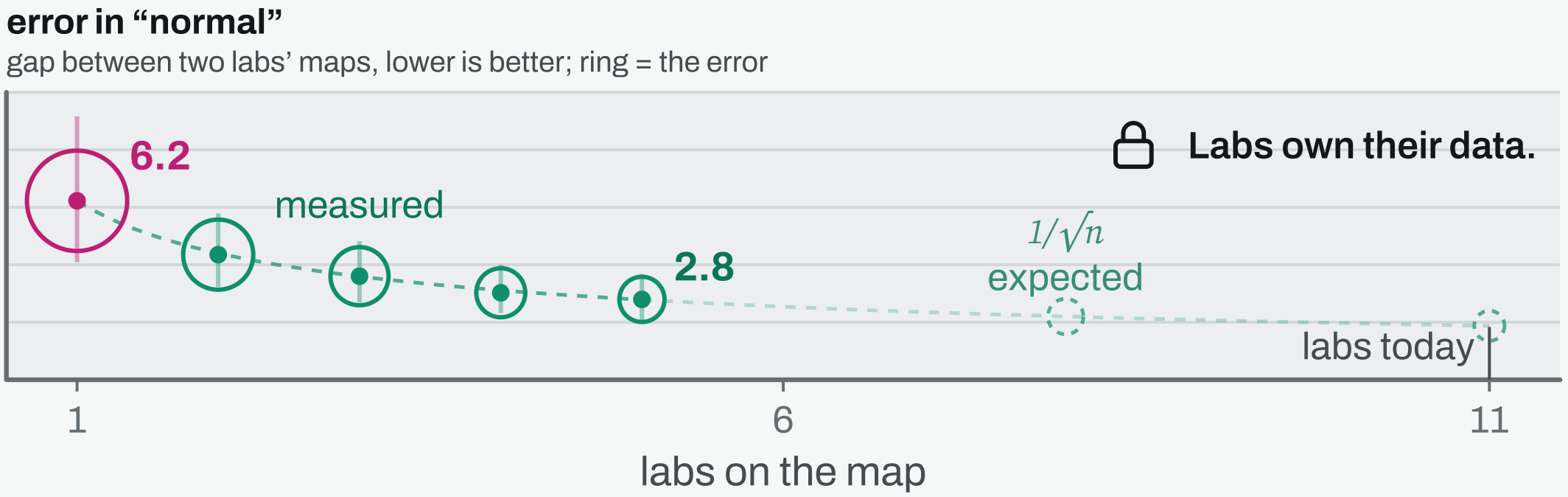






93,196 of the 93,228 wells on the map (one plate has no batch). Public archives keep finished datasets (656 TB, free [17]); this keeps every lab's live controls, aligned and checked. Left: AMG900, JUMP's strongest positive control, an Aurora kinase inhibitor; per lab, one plate from each of up to three batches whose known answer came out, site 1; each crop is one whole cell, the one of median nuclear area, 90 µm across, the rest black; one brightness scale per lab from its DMSO cells, for both conditions; DNA (Hoechst) pink, AGP (phalloidin, WGA) green; bars 20 µm; dashed: mean change; 5 µM, JUMP [7] and JUMP-Target's recommended dose (results/e18). Right: mean agreement across labs on each known drug's change (cosine, 5 µM), each batch read against its own controls (alone) or on the shared map (aligned); 125 batches, 11 labs (results/e12, A4). JUMP [7], A5, A6, A11

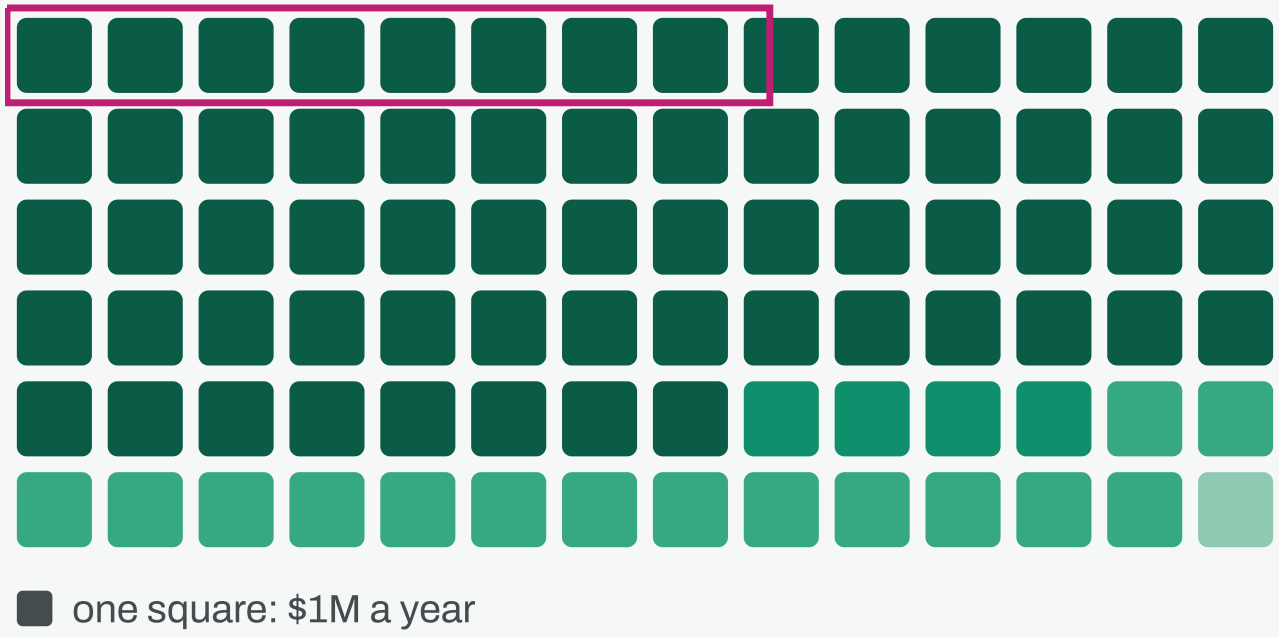
Every lab that joins sharpens “normal” for everyone.



Why now 2023 1.8B cells, public [7, 25] 2025 FDA: fewer animal tests [9] 2025 $30M for one cell map [11] now AI

fig 5 error: distance between two maps from different labs, in within-plate spreads; dot: mean of 400 draws, bar: 10th to 90th percentile; measured to five labs, dashed beyond; 1,871 plates; JUMP [7]; results/e13

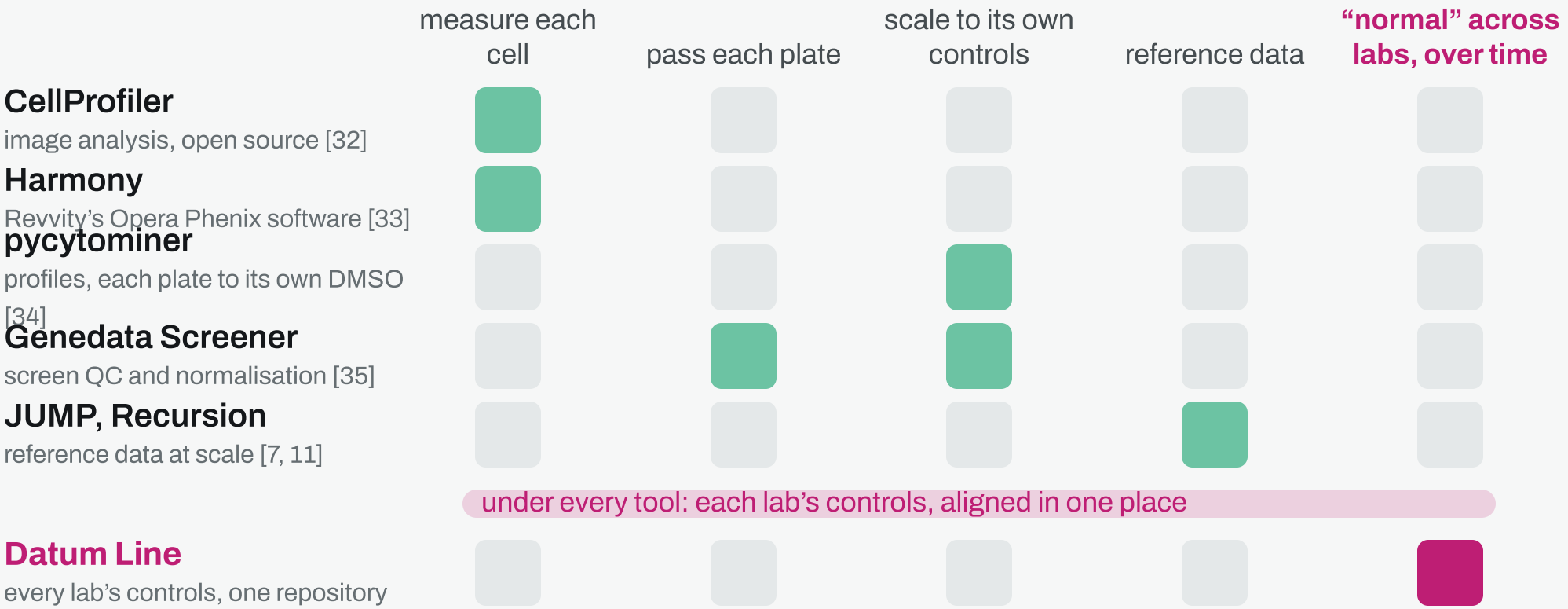
Every likely buyer: $84M a year.
Five-year target: a tenth.



screening labs **$10k a line, a quarter**	**$64M** 400 labs × 4 lines
big screening panels **$150k a year**	**$3.75M** 25 panels
buyers of cell data **$25k per dataset**	**$15M** 150 buyers × 4 datasets
benchmark partners **$100k a year**	**$1M** 10 partners
academic labs they share controls; they are the map	**free**
AI teams priced by use, not yet set; not counted	**by API**

fig 6 base case of A10: every count and price an assumption until a count replaces it; low case $9.7M a year; academic labs join free and are the map, not the revenue; AI usage by API not priced; why now is on 06

Each tool sees one lab.
Datum Line sees every lab.



tools as phenotypic screen suppliers: CellProfiler [32]; Harmony, Revvity [33]; pycytominer [34]; Genedata Screener [35]; JUMP [7], Recursion [11]. Each works inside one lab’s data; none compares a lab’s controls with other labs’. The wider landscape (cell-line checks, vendor certificates): docs/landscape.html

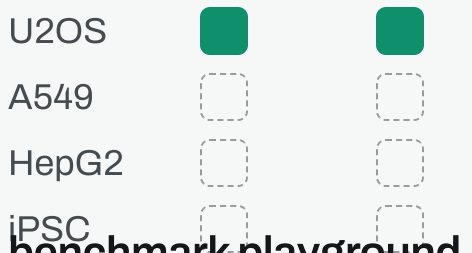
landscape ↗

Raising $1.8M pre-seed:
18 months to first revenue.

people	$1.25M	benchmark	$0.30M		$0.22M
a leader to run it	$360k	playground, ~60 plates	$240k	compute	$90k
scientific lead (me)	$360k			legal, IP	$90k
ML engineer, from month 7	$320k	3 partner labs	$60k	travel	$40k
payroll overhead	$208k				

$884k planned × 2 for overruns (A9)

1 → 2 → 3 → 4 → seed round



benchmark playground

known drugs, planted faults, archive A4, archive A16

11 + 3



three partner labs on the map

labs + 3 partners

U2OS first, then any line

99.5%



model for any image

right lab of 3 labs with brightfield

from one brightfield image; chance 12.5%, A3, A7



first paid certificate

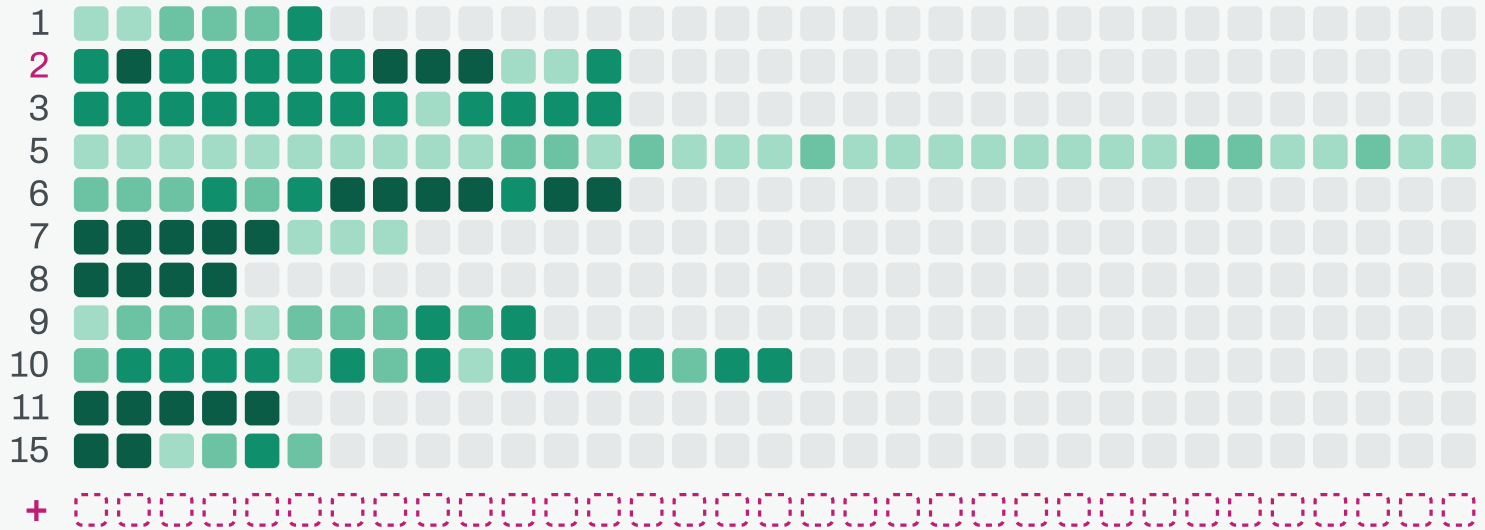
a company pays ~$10–50k per check, A5

proposed, milestones in order, not dated

roadmap ↗

Every plate imaged, placed on the map.

Labs share control images, free.
Each sees its “normal” next to everyone else’s.



129 batches today; the open row, yours

Less More plates

$1.8M

pre-seed:
18 months to first revenue, A9

LOOKING FOR

- **1** leader, to run it
- **3** partner labs, bench time
- **2** benchmark partners

Have control images? Talk to me.

fig 7 one square: one batch, in run order, shaded by plates at the quartiles; a median of 10 plates (1 to 58); 1,871 plates, JUMP [7], results/e9; A5, A8, A10, A11

all documents ↗

The rest, for questions.

Every number: the code here or a cited source.
All 25 boards: archive.html
Documents: white paper, product vision, roadmap, landscape

A1 Why not batch-correct it away?
erases most of the lab fingerprint

A2 Does it change the result?
same drug, different answer

A3 Does it work without stains?
bet on brightfield

A4 What does the map buy?
same drug, closer between labs

A5 Who pays?
labs join free, subscribers pay

A6 What is private?
control wells leave, nothing else does

A7 What is the machine learning?
image in, position and correction out

A8 What about biosecurity?
cells show their lab, labels can lie

A9 How much, and what for?
planned $0.9M × 2 = $1.8M asked

A10 How big?
count × price, three streams

A11 Why would a lab share?
owned by the lab, verified by anyone

The usual correction erases most of the lab's fingerprint.

JUMP records the left column; nobody records the middle.

RECORDED IN JUMP

- lab
- batch date
- plate
- plate type
- well position
- microscope
- confocal or widefield
- laser or LED
- objective NA
- fields per well
- filter set
- pixel size
- brightfield planes
- CellProfiler version

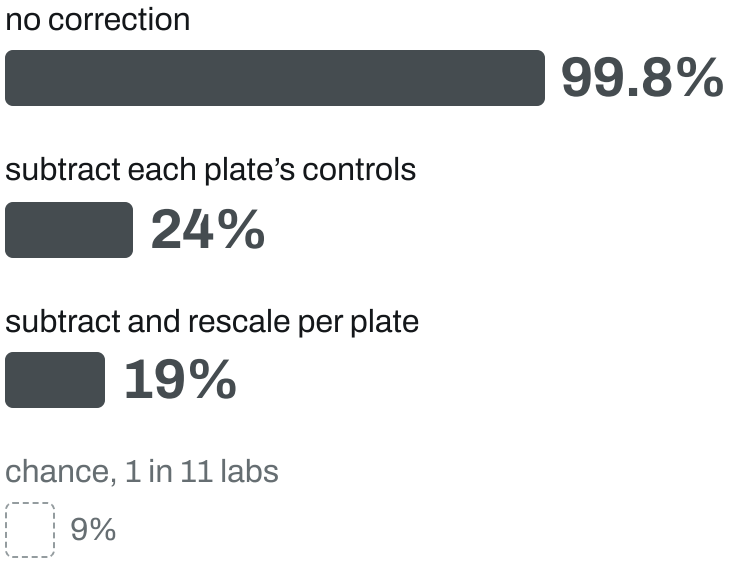
in every image file too:
image size, stored bit depth

NOT IN ITS PUBLIC METADATA

- DMSO % per well
- DMSO lot
- DMSO water uptake
- stock freeze–thaw
- serum lot
- media lot
- passage
- thaw date
- seeding density
- confluence at dosing
- dosing to fixation
- stain and dye lots
- operator
- incubator shelf
- edge evaporation
- mycoplasma status
- CO2 and humidity
- compound dose

not required to publish today
in our own data: tracked, every one

CAN A MODEL TELL WHICH LAB?
HIGHER = CLEARER LAB FINGERPRINT



29,943 DMSO wells, 16 per plate
tested on plates it never saw

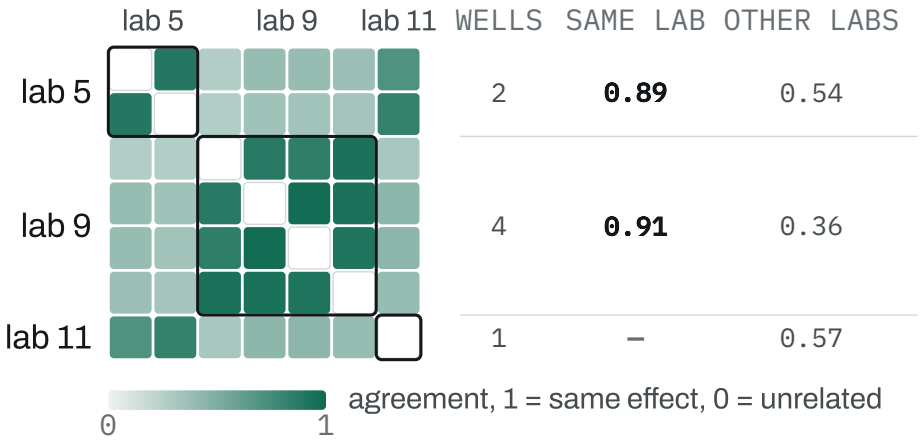
Batch correction is built to remove this signal [18], variables: JUMP metadata [7]

The same drug looks less alike in another lab.

Darker = more alike.

WHAT A DRUG SCREEN REPORTS — drug wells – DMSO wells, same plate

RSL3 AT 10 µM, EVERY WELL AGAINST EVERY OTHER



	WELLS	SAME LAB	OTHER LABS
lab 5	2	**0.89**	0.54
lab 9	4	**0.91**	0.36
lab 11	1	–	0.57

SAME 7 PLATES, EACH WELL AGAINST ITS OWN PLATE'S DMSO

	SAME LAB	OTHER LABS
RSL3, 10 µM	**0.90**	0.49
8 known drugs, 5 µM, median	**0.74**	0.26
DMSO vs DMSO	**0.00**	0.00

DOES IT CHANGE THE NUMBER?

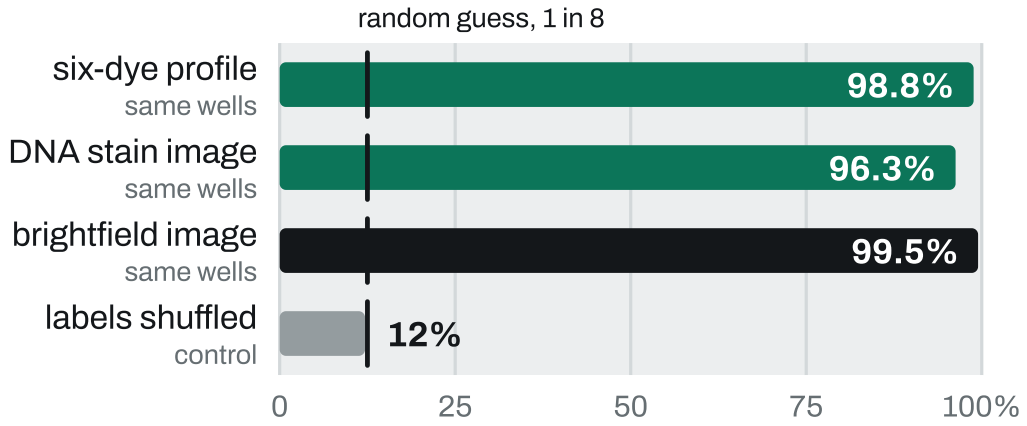
Which lab ran it explains 20% to 71% of how strong a drug's effect looks.

Seven wells: an example, not proof, lab 7, dosed at a sixteenth, left out, agreement = cosine of two wells' effects, each well minus its own plate's DMSO wells, 8 drugs at full scale: 1,829 plates, 11 labs, JUMP [7]

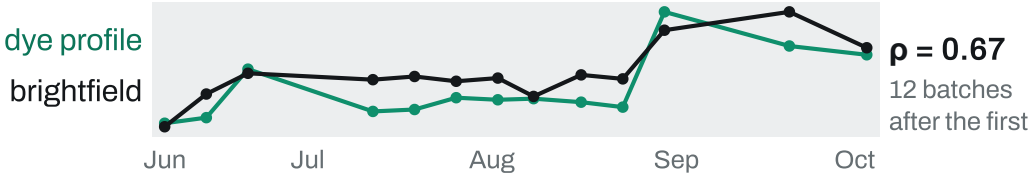
One brightfield image already names the lab.

Evidence: six dyes.
Product: no dye.

CAN ONE WELL TELL WHICH OF 8 LABS IT CAME FROM?



DOES BRIGHTFIELD SEE LAB 2 DRIFT?, EACH LINE SCALED TO ITS PEAK

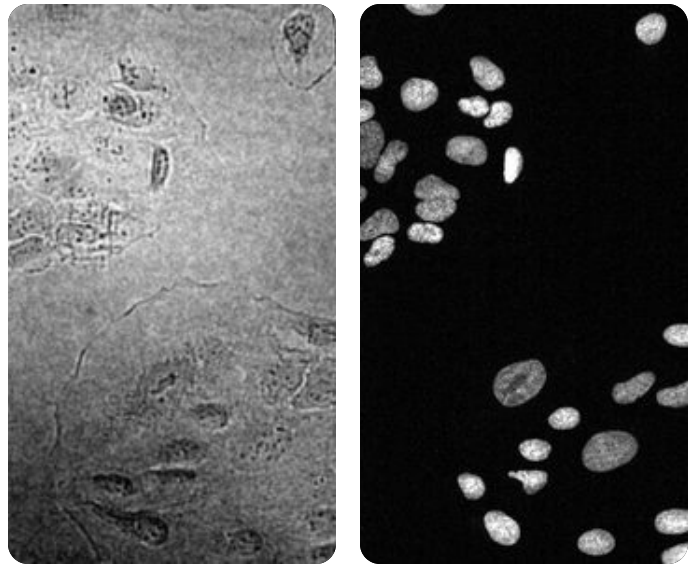


Not yet the whole map: beyond each lab's usual neighbour, brightfield and dye profile agree batch by batch no more than luck (45 of 102 against 41).

WHY THE BET IS FAIR

No dye: fewer things can go wrong

~~stain lot~~ ~~dye lot~~ ~~staining step~~ A1



BRIGHTFIELD DNA STAIN

same spot in one well, lab 2

% named right, each lab counted equally, unseen batches, one DMSO well per plate, 1,338 plates, the same wells and model for every row, median of 20 splits of the batches, profile without its 8 frame-size columns, shuffled: 5 seeds per input, image model never trained on cells, JUMP [7]

On the map, labs agree a little more on the same drug.

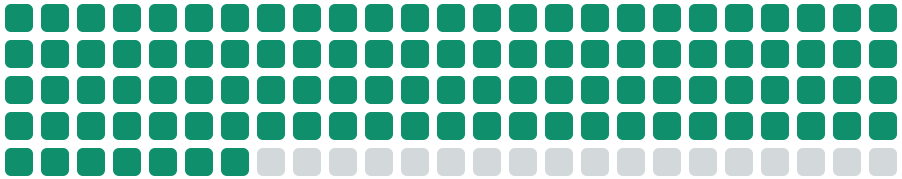
Seven known drugs, 125 batches.

Do labs agree on what a drug does?

0.75 → **0.76** → **0.78**

raw | usual fix | on the map

1 = every lab sees the same effect



107 of 125 batches higher on the map than with the usual fix

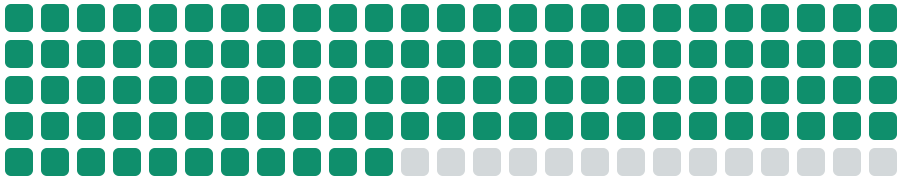
than raw: 121 of 125

Can you still tell the drugs apart?

0.52 → **0.56** → **0.59**

raw | usual fix | on the map

gap to the nearest wrong drug, higher is easier



111 of 125 batches higher on the map than with the usual fix

than raw: 125 of 125

NOTHING LOST

Right drug recognised: 0.92 → 0.94

Matches its own past: 0.88 → 0.88

HOLDS

Labs agree more: 5 of 5 ways of checking

Drugs stay apart: 4 of 5

NOT FIXED YET, DRUG STRENGTH

Varies batch to batch, lower is better

raw 6% → usual fix 20% → map 12%: steadier than the usual fix, not yet as steady as raw

usual fix: each batch set against its own “normal”, its DMSO wells, map: against every other lab’s, 1,829 plates, JUMP [7]

Labs earn access by uploading; AI teams pay the most, by API.

WHO	WHAT THEY GET	WHAT THEY PAY	WHAT IT BUILDS
labs and core facilities	access to the repository and the map; certificate, leaderboard	**nothing: uploading controls earns their access**	the map
AI labs and model builders	certified raw images from every lab, by API	**the most: priced by use, not yet set**	main revenue
companies that buy cell data	a certificate on every dataset they buy	**~$10–50k per certificate**	first revenue
teams that run cell screens all year	a check on every batch they run	**~$5–15k per cell line, per quarter**	repeat revenue
benchmark partners	a test set with faults planted on purpose	**~$50–200k a year**	the standard

only control wells leave a lab, DMSO and known drugs, never a test compound, privacy A6, the lab’s share A11

Only control wells leave the lab.

Nothing for legal to approve.

WHAT LEAVES THE BUILDING

control wells only: DMSO and known drugs



green: 64 of 384 wells leave, JUMP [7]

images, or numbers from them

proposed: images deleted once measured

a plate map and the batch date

which wells, and when they were imaged

WHAT DATUM LINE DOES WITH IT

builds the map

scale set by the other labs, never its own

the map holds DMSO wells only

82,319

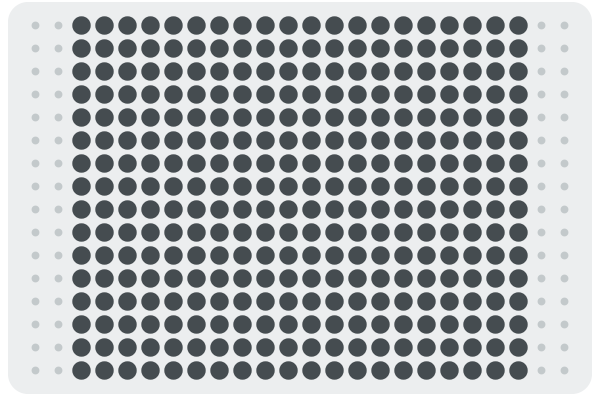
for lab 2: DMSO wells of 10 other labs
0 treated

answers

“normal” or not, how far it moved, nearest lab

WHAT IT NEVER SEES

your treated wells



dark: 320 of 384 wells stay with you

your results

corrected in your building, by you

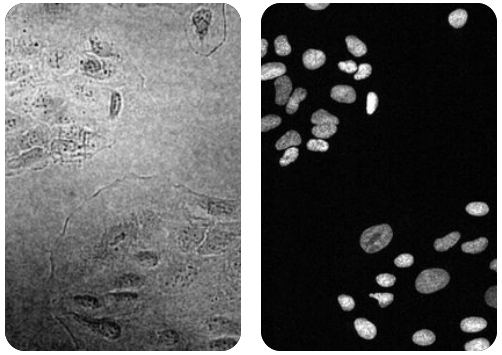
your name, unless you choose

proposed: a number until you opt in.
Every lab here is a number.

In every analysis here, the map and the correction see DMSO wells only, github.com/kat-titter/datum-line/analysis, JUMP [7]

An image goes in; a position and a correction come out.

1, AN IMAGE



brightfield DNA stain

one DMSO well, JUMP [7]

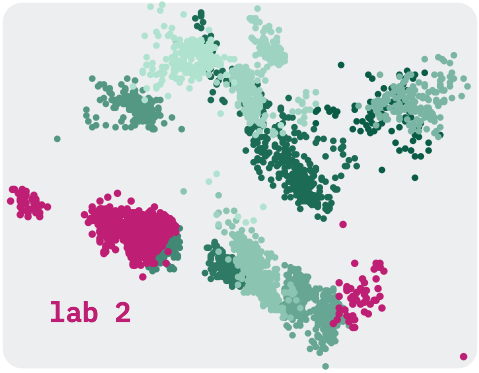
2, NUMBERS

3,773

numbers per well

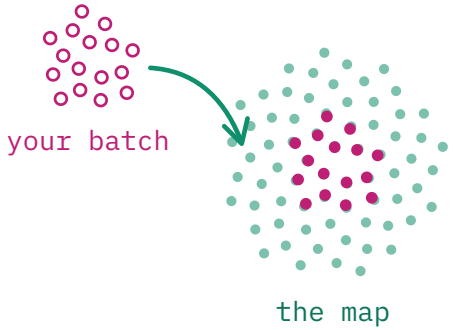
standard software today: size, shape, brightness, texture

3, THE MAP



each lab placed on the other labs' scale, never its own

4, THE CORRECTION



one per batch, fitted on your DMSO wells, sketch, A4

AI IMAGE MODEL NEXT, A3



right lab, from one picture

Correction, per batch on 200 components: $x' = C_{field}^{\frac{1}{2}} C_{batch}^{-\frac{1}{2}} (x - m_{batch})$, every script: github.com/kat-titter/datum-line/analysis

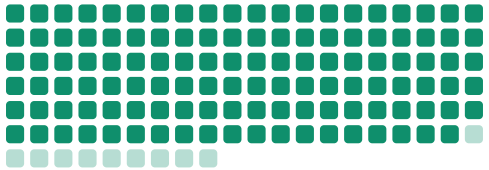
Every lab leaves a fingerprint, so a wrong label stands out.

Two work today, three are proposed.

PROVENANCE today

99.9%

of 93,228 DMSO wells get the right lab, on unseen plates: cells and microscope together.



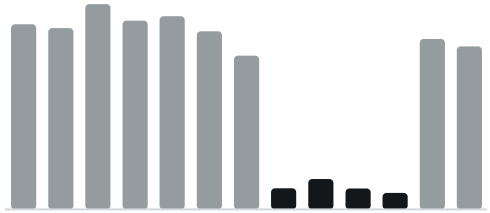
1 square = 1 batch
solid: every well right, 119

INTEGRITY today

4 batches

Label: known drug in these wells.

Cells: almost no effect, on 65 plates.



lab 3, drug effect per batch
dark: failed, archive A8

UNKNOWN ORIGIN proposed

9×

Farthest of 129 real batches from its nearest other lab.

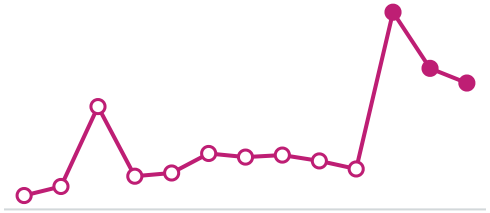


distance to nearest other lab
1 line = 1 real batch

EARLY WARNING proposed

30 Aug

2021: lab 2's DMSO wells moved; its cell count did not.



lab 2, since batch 1
filled: flagged, archive A13

DECLARED VS OBSERVED

proposed, opt-in

A company that wants the good-actor mark sends treated wells and says what is in them. The map flags cells that look like a known toxic class [7]. Only what a lab sends is seen.

Integrity check uses each plate's known-drug wells (archive A8), × = times the usual spread of wells within one plate, JUMP [7]

proposed, every line an assumption

About $1.8M for eighteen months, mostly people.

$884k planned × 2 = $1.77M asked



people 71% | benchmark 17% | other 12%

leader, to run it	18 months	**$180k**
scientific lead (me)	18 months	**$180k**
ML engineer	from month 7	**$160k**
payroll overhead	20% of salaries	**$104k**
benchmark playground	~60 plates, planted faults, archive A16	**$120k**
partner-lab bench time	3 labs, controls they already image, reagent top-ups, shipping	**$30k**
compute and storage	images, models, hosting, API	**$45k**
legal, IP, insurance, accounting	incorporation to first contract	**$45k**
travel and partner visits	2 benchmark partners, 3 labs	**$20k**
planned	sum	**$884k**
+ buffer	most things take 2–3× the time and money planned, we plan for 2×	$884k
asked, 18 months		**$1.77M**

NO IN-HOUSE LAB

Partner labs or a contract lab.
Own lab: costs more, proves less.

WHAT THE MONEY MUST SHOW

1 benchmark playground, paid in this round **2** three partner labs on the map
3 model for any image **4** first paid certificate → seed

salaries below market, leader and scientific lead paid, nothing here committed

About $84M a year, if every likely buyer joined.

assumption	low	base	high
organisations testing drugs on a few cell lines	200	400	800
cell lines each one uses per quarter	2	4	8
price per cell line per quarter, $k	5	10	15
organisations testing drugs on hundreds to thousands of cell lines	10	25	50
site licence per year, $k	50	150	300
per-line checks and site licences, per year	**$8.5M**	**$67.8M**	**$399M**
companies buying cell data they did not make	50	150	300
purchased datasets each certifies per year	2	4	8
price per certificate, $k	10	25	50
certificates on purchased data, per year	**$1M**	**$15M**	**$120M**
benchmark partners	4	10	20
membership per partner per year, $k	50	100	200
benchmark membership, per year	**$200k**	**$1M**	**$4M**
total per year, if every buyer joined	**$9.7M**	**$83.8M**	**$523M**

BASE CASE, PER YEAR



↑ five-year target, a tenth, $8.4M

benchmark $1M ↑

WHY BUYERS PAY

A line checked late, in a screen I ran:
3 runs a week × 17 weeks × $50k ≈ $2.6M to redo.
A $25k certificate: about 1% of that.

prices proposed on A5, academic labs join free, they are the map, not the revenue

The lab keeps its data and a share; anyone can check it.

No crypto token.

CONTROL WELLS ONLY — today

no test compound, no target, no result
results never leave the lab

LAB KEEPS ITS DATA — proposed

sharing is opt-in
withdraw any time

LOW RISK BY DESIGN — today

cell-line images only
no human genetic data, no pathogen data

SIGNED AND TIMESTAMPED — proposed

public record, anyone can check
dates cannot be changed, blockchain optional

KNOWN SUBSCRIBERS — proposed

verified at sign-up, odd access flagged
access log open to members

lab shares
images of control cells
certificate attached

→

certificate checks
how “normal” the cells look
which lab, what changed

→

subscriber pays
to use certified data
from many labs

→

lab keeps a share
of every subscription
as long as its data is used